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Composition design of high entropy alloys using the valence electron concentration to balance strength and ductility

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ABSTRACT

The valence electron concentration (VEC) is an important physical factor for phase formation. A high VEC is conducive to forming an FCC phase and improving an alloy's ductility, while a low VEC is beneficial in forming a BCC phase that improves an alloy's strength. This is demonstrated for two HEAs, CoCrCuFeNi (FCC) and AlCoCrFeNi (BCC), that were designed as matrix alloys, where Ni and Mo are alloyed. The microstructure, phase evolution, and the mechanical properties for $(AlCoCrFeNi)_{100-x}Ni_x$ and $(CoCrCuFeNi)_{100-x}Mo_x$ were systematically investigated. As the phase structure for the $(AlCoCrFeNi)_{100-x}Ni_x$ high entropy alloy (HEA) transformed from a BCC to an FCC crystal structure as the Ni content increased from 0 at.% to 16 at.%, the FCC volume fraction increased from 0% to 85%, its compressive fracture strain increased from 25% to 40%, its VEC increased from 7.2 to 7.6. As the phase structure for the $(CoCrCuFeNi)_{100-x}Mo_x$ HEA transformed from FCC to BCC as the Mo content increased from 0 at.% to 16 at.%, the BCC volume fraction increased from 0% to 65%, its compressive yield strength increased from 260 MPa to 928 MPa, its VEC decreased from 8.8 to 8.3. Selecting an element based upon an alloy's VEC is a practical method for designing compositions for HEAs that balance strength and ductility. According to the needs of practical applications, balancing both strength and plasticity requires the following criteria for selecting an element for incorporation into an HEA system: matrix strength is improved by selecting an element with a VEC lower than the average VEC of the matrix, while ductility is improved by selecting another element with a VEC higher than the average VEC for the matrix.

1. Introduction

Biological triumphs include the double-helix, the genetic code, and the synthesis of artificial cells. Is there a corresponding materials genome [1]? High-entropy alloys (HEAs), with their multiple principal elements, have attracted extensive research attention [2–5] since the concept was independently proposed in 2004 by Yeh [6] and Cantor [7]. HEAs provide the possibility for numerous combinations of novel compositions and properties [1,3,4], which highlights the importance of alloy design in HEA systems.

Recently, efforts utilizing scientific criterion and approaches for phase prediction were applied in HEA-related topics, such as the entropy of mixing ΔS_{mix} [4,6], atomic size differences δ and melting points T_m [8], the enthalpy of mixing ΔH_{mix} [9], the valence electron concentration (VEC) [10,11], and the scale ratio of ΔS_{mix} to ΔH_{mix}, which is represented by Ω [8]. According to previous researches [8,9], if two parameters satisfy the conditions $\Omega \geq 1.1$ and $\delta \leq 6.6\%$, then a stable, solid-solution phase is formed in an HEA system [8,9]. When the VEC≥8, then the face-centered cubic (FCC) structure's solid-solution phase is formed. But when $6.87 \leq$ VEC<8, both the FCC and body-centered cubic (BCC) structure's solid-solution phases are formed. And when the VEC<6.87, only a single BCC, solid-solution phase is present [10,11]. While this is helpful in some aspects of alloy design, more quantitative predictions are still far from complete and requires significantly more experimentation to understand them. Traditional alloy design involves selecting the major elemental component based upon a specific property requirement, and then using additional elements to confer secondary properties without sacrificing the primary property [12]. Several similar methods have been used in designing HEAs [13–20]. For example, Al, V, and Mo were added to FeCoNiCrMn

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and CoCrFeNi to study the effects of different elements on the structural evolution and mechanical properties of these HEA systems [12,21,22]. As the Al, V, and Mo concentrations increased, the crystalline structure changed from the initial, single ductile, FCC phase to the combined, hard BCC/HCP structure, for FeCoNiCrMn and CoCrFeNi respectively. These results imply that some metal elements have the potential to directly change phase structures and the mechanical properties in HEA systems. HEAs possessing ductile, FCC solid solutions have greater plasticity and lower strength, because they have an abundance of slip directions in the crystal structure [23]. On the other hand, HEAs with hard, BCC/HCP solid solutions are stronger with lower plasticity, because these crystal structures possess fewer slip directions [24]. Meanwhile, single-phase, HEAs could not easily achieve a reasonable balance between strength and plasticity, but a composite FCC and BCC/HCP phase is expected to achieve this balance [25]. The design of HEAs with dual phases – that are both ductile and hard – is an urgent problem in the composition design of HEAs, especially given the existence of limited HEA designs [3].

In this paper, we introduce an alloy design method based on the relationship between the VEC and the phase structure. We can control the phase structure by adjusting the proportion of each element to optimize the mechanical properties of HEAs. To verify the validity of this alloy design method, two classical HEAs, CoCrCuFeNi and AlCoCrFeNi with possess respective FCC and BCC structures, were selected as matrix alloys for this investigation. A systematic alloying study with Mo and Ni is conducted by investigating the microstructure, phase evolution, and the mechanical properties of $(CoCrCuFeNi)_{100-x}Mo_x$ and $(AlCoCrFeNi)_{100-x}Ni_x$. In addition, atomic diffusion models were proposed to describe the phase-forming process, and the relationship between the microstructure and the mechanical properties are discussed.

2. Theoretical analysis and the experimental method

Alloy elements are divided into two categories: one type consists of compact elements or C-elements, and another type is comprised of loose elements or L-elements. C-elements reduce the distance between the C-element and neighboring atoms in the matrix, where the C-element possesses a larger VEC than the average VEC for the matrix. The high atomic density favors the FCC phase. Meanwhile, L-elements increase the distance between the L-element and neighboring atoms in the matrix, where L-elements possess a lower VEC than the average VEC for the matrix. The lower atomic density promotes the BCC phase. Fig. 1 shows a schematic diagram that demonstrates the VEC effect on the atomic arrangement. Fig. 1(a) illustrates the atomic arrangements for matrix alloys, where f_1 represents the interactions between two atoms in the matrix, and d_1 represents the distance between these two atoms. The addition of C-elements, which possess a larger VEC than the average VEC for the matrix, are incorporated into the matrix. Fig. 1(b) shows the interactions, between a C-element and its neighboring atoms in the matrix, are increased upon the C-element's introduction into the matrix. The increased force is represented by f_2, while the reduced distance between these two atoms are denoted by d_2. The values in Fig. 1(b) may be compared to their initial values as $f_2>f_1$ and $d_2<d_1$. In the opposite case, where the VEC is below the average VEC of the matrix, interactions between an L-element and its neighboring atoms in the matrix, f_3, will be reduced after an L-element is introduced into the matrix. The distance, d_3, will be increased, as shown in Fig. 1(c), compared to the initial matrix shown in Fig. 1(a).

Fig. 1(d) shows the FCC cell structure and its slip systems, while Fig. 1(e) illustrates the BCC cell structure and its corresponding slip systems. The FCC cell structure has a higher atomic density than the BCC cell structure, while the distance between two atoms are also smaller than for the BCC cell structure. Therefore, when atoms are added into an HEA system, and possess a larger VEC than the average VEC for the matrix, then the FCC phase is favored. Conversely, when atoms with a lower VEC than average VEC for the matrix are added into an HEA system, the BCC phase is encouraged. HEAs with ductile, FCC solid solutions have better plasticity and lower strength, because the FCC crystal structure possesses more slip directions than the BCC structure [23]. HEAs with hard, BCC solid solutions, however, are stronger and demonstrate lower plasticity compared to FCC lattices, because its crystal structure contains fewer slip directions [24]. Naturally, a composite containing both BCC and FCC phases are expected to achieve this balance [25]. According to the needs of practical applications, balancing both strength and plasticity requires the following criteria for selecting an element for incorporation into an HEA system: matrix strength is improved by selecting an element with a VEC lower than the average VEC of the matrix, while plasticity is improved by selecting another element with a VEC higher than the average VEC for the matrix.

Table 1 shows the VEC for Mo and Ni elements, and the average VEC values for CoCrCuFeNi and AlCoCrFeNi. It shows that Mo has a lower VEC than the average VEC in CoCrCuFeNi, while Ni has a VEC higher than the average VEC in AlCoCrFeNi. The CoCrCuFeNi HEA demonstrates ductile, FCC solid solutions with improved plasticity and a lower strength due to the existence of more slip directions in its crystal cell. Meanwhile, the AlCoCrFeNi HEA possesses hard, BCC solid solutions that are stronger, but with limited plasticity, which stems from this crystal structure containing fewer slip directions. To further prove this methodology's feasibility for designing alloys with customized properties, Mo, which possesses a lower VEC than the average VEC value for CoCrCuFeNi, was selected as the alloy element for incorporation into the CoCrCuFeNi HEA matrix. The aim is to improve this HEA's strength by generating a hard, BCC phase in the alloy. Additionally, Ni, which has a larger VEC than the average VEC value in AlCoCrFeNi, was added into the AlCoCrFeNi HEA to improve the structure's plasticity by generating a ductile, FCC phase in this alloy.

A series of $(CoCrCuFeNi)_{100-x}Mo_x$ and $(AlCoCrFeNi)_{100-x}Ni_x$ HEAs, where $x = 0$, 4, 8, 12, and 16, were prepared by arc-melting a mixture of pure metals (purity˃99%) in a high-purity, argon atmosphere. They were melted seven times to improve the chemical homogeneity. Samples were prepared for microstructure analysis by first polishing use 2000 sandpaper, and then electro-polishing the sample in a mixture of 90% acetic acid and 10% perchloric acid at room temperature. The HEA phase structures were identified by X-ray diffraction (XRD) using Cu K_α radiation (MXP21VAHF), while their microstructures were characterized by scanning electron microscopy (SEM). Compression engineering stress-strain curves were recorded using an initial strain rate of 1×10^{-3} min^{-1} at room temperature with an INSTRON Universal Tester. For each specimen, at least 3 samples were measured to confirm reproducibility. Strain was measure by LVDT displacement transducer. The yield strength and the strain were derived directly from the stress-strain curves.

3. Results

Fig. 2 shows the XRD patterns for $(AlCoCrFeNi)_{100-x}Ni_x$ and $(CoCrCuFeNi)_{100-x}Mo_x$. As the Ni and Mo content were increased, the XRD patterns exhibited an intended, structural phase transformation from FCC to BCC or from BCC to FCC (refer to the X-ray diffraction information in Fig. 2). Adding both Ni and Mo enabled the HEA systems to attain a dual-phase microstructure, where both phases maximized the strengthening effect in the solid-solution

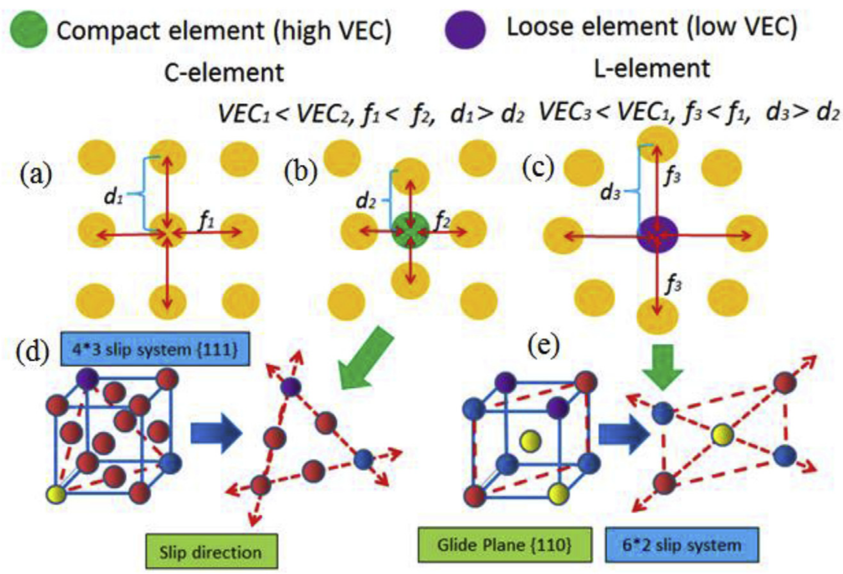

Fig. 1. The effect of the VEC on the atomic arrangement. (a) A representative atomic arrangement for matrix alloys, where f_1 and d_2 respectively represent the force and distance between two atoms in the matrix. (b) The change in the local force, f_2, and distance, d_2, between an atom and a C-element with a larger VEC than the average atomic arrangement. (c) An L-element with a lower VEC on atomic arrangement will create a different force, f_3, and distance, d_2, between an atom and the L-element. (d) The slip systems and the atomic arrangement for an FCC crystal structure. (e) The slip systems and the atomic arrangement for a BCC crystal structure.

Table 1
The valence electron concentrations (VECs) obtained from Ref. [11].

Element	Mo	Ni	CoCrCuFeNi	AlCoCrFeNi
VEC	6	10	8.8	7.2

compared to the presence of only a single phase [26].

Fig. 2(a) shows the XRD patterns for $(AlCoCrFeNi)_{100\text{-}x}Ni_x$. When the Ni content was between 0 and 4 at.%, a single, BCC phase structure was exhibited. A dual-phase, FCC/BCC structure was achieved when the Ni content was between 8 and 16 at.%. The peak intensities of FCC (225) also increase at the same time, which indicates the FCC volume fraction increase when Ni content increases from 8 to 16 at.%, it has a positive effect in improving the fracture strain. The XRD patterns for $(CoCrCuMnNi)_{100\text{-}x}Mo_x$ are displayed in Fig. 2(b). When the Mo content was between 0 and 4 at.%, only the FCC phase was detected in this HEA system. Meanwhile, a dual-phase, FCC/BCC structure was achieved when the Mo content was between 8 and 16 at.%. The peak intensities of BCC (229) also increase at the same time, which indicates the BCC volume fraction increase when Mo content increases from 8 to 16 at.%, it has a positive effect in improving the yield strength. These results imply that Ni and Mo induced phase transformation processes.

Fig. 3(a) plots the elemental distribution maps for $(AlCoCrFeNi)_{84}Ni_{16}$ HEA. The FCC phases were Al-poor phase. Based upon the elemental distribution maps for $(CoCrCuMnNi)_{84}Mo_{16}$ HEA as shown in Fig. 3(b), Mo-rich, BCC phases were detected. This is also indirect proof that Ni and Mo induced the phase transformations.

Fig. 4 displays microstructures for the as-cast, HEAs with different Ni contents, which are shown in the left column. The regions labelled in the left column are magnified in the right column. Fig. 4(a) revealed that the AlCoCrFeNi HEA possessed a single-phase structure. The high magnification image, labelled as region A in Fig. 4(a), appears in Fig. 4(b), where honeycomb-like structures were present with a uniform dispersion of spherical particles. These structures were similar to the structures reported in the AlCoCrFeNi HEA prepared by Bridgman solidification [27], and the as-cast, Fe-Ni-Mn-Al alloy system [28]. Fig. 4(c) contains microstructures for the $(AlCoCrFeNi)_{96}Ni_4$ HEA, which revealed the presence of Al-poor, FCC phases present in the grains boundary, and as observed in Fig. 4(d), they were also present in the inner grains, which were not detected by XRD due to their small volume fraction. Fig. 4(e) and (f) shows the microstructures for the $(AlCoCrFeNi)_{92}Ni_8$ HEA, Al-poor, FCC phases were founded in the $(AlCoCrFeNi)_{96}Ni_4$ HEA, and they are larger than the precipitated in $(AlCoCrFeNi)_{96}Ni_4$ HEA. Fig. 4(g-j) respectively display microstructures for $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEA, where $x = 12$ and 16. These micrographs indicated that the precipitated, Al-poor, FCC phases became gradually larger as the Ni content increased. It also suggested that Ni has a positive effect on the phase transformation from BCC to FCC.

The black areas, as shown in Fig. 4(g-i), were identified as oxygen with EDS, and were previously reported in other HEAs [29–32]. It was uncertain if the oxygen was introduced through the raw material or during one of the subsequent processing steps. Because of their tiny volume fraction, the oxide particles did not significantly influence the HEA's mechanical properties.

To more conclusively prove the feasibility of the alloy design method proposed herein, Mo, with has a lower VEC than the average VEC value for the CoCrCuFeNi matrix, was also selected as an element for incorporation into the CoCrCuFeNi matrix. Fig. 5 shows the microstructures for this as-cast, HEA with different Mo concentrations.

Fig. 5(a) indicated that the CoCrCuFeNi HEA possessed a single-phase structure. The high magnification image in Fig. 5(b), which is denoted as region A in Fig. 5(a), revealed Cu-rich phases located in the grain boundaries shown in Fig. 5(b-d) show corresponding low and high magnification of microstructures for the $(CoCrCuFeNi)_{96}Mo_4$ HEA, where Cu-rich phases were founded inside the grains. Fig. 4(e) and (f) respectively displays low and high magnification images for microstructures belonging to the $(CoCrCuFeNi)_{92}Mo_8$ HEA, where Mo-containing, BCC phases were generated, and as shown in Fig. 5(g-j), these phases appeared to increase the volume fraction as the Mo concentration increased. These results suggested that Mo had a positive effect on the phase transformation from FCC to BCC.

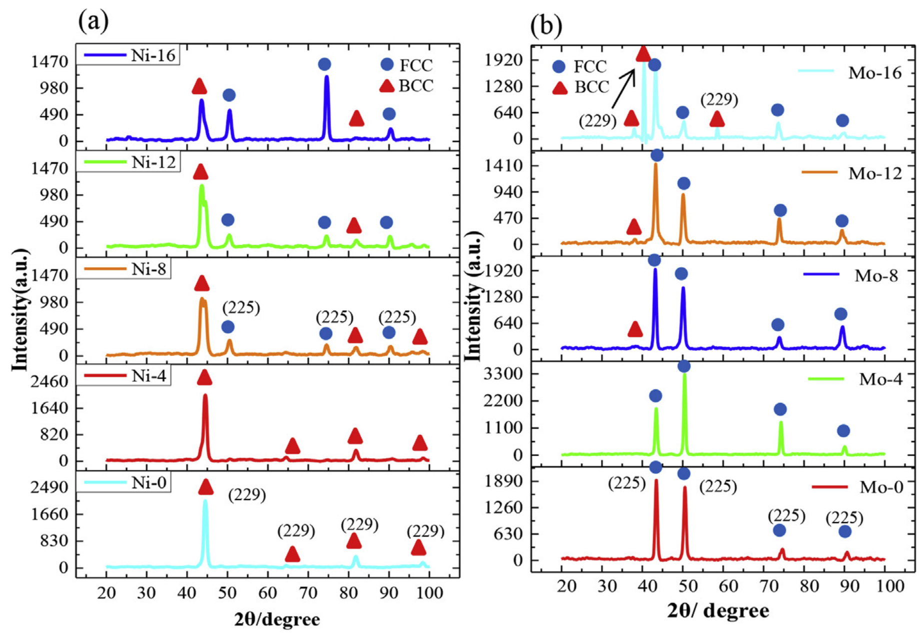

Fig. 2. XRD patterns for (a) $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ and (b) $(CoCrCuFeNi)_{100\text{-}x}Mo_x$.

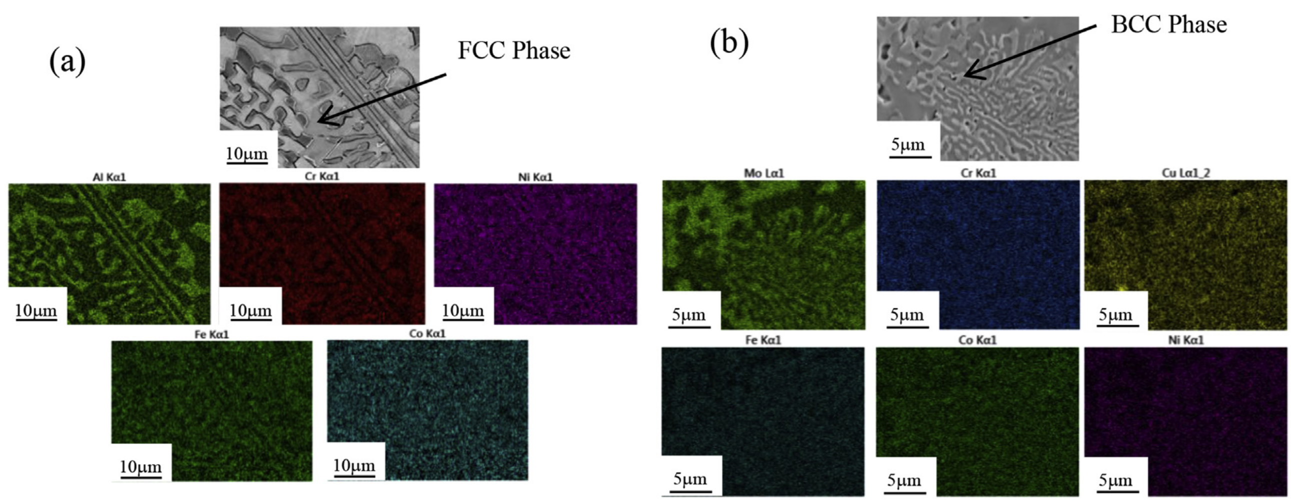

Fig. 3. Elemental distribution maps for dual phase (a) $(AlCoCrFeNi)_{84}Ni_{16}$ and (b) $(CoCrCuFeNi)_{84}Mo_{16}$.

Fig. 6 shows compressive mechanical properties for the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ and $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEAs. The following changes occurred in the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEA as the Ni concentration was increased from 0 at.% to 16 at.% (refer to Fig. 6(a)): The fracture strain increased from 25% to 40%, the compressive yield strength changed from 1245 MPa to 355 MPa, and the ultimate strength decreased from 3092 MPa to 2639 MPa. As the Mo concentration increased from 0 to 16 at.%. In the $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEA, the compressive yield strength increased from 260 MPa to 928 MPa, but the fracture strain decreased from 60% (no fracture) to 16% (refer to Fig. 6(b)). These results agree well with the strength-ductility trade-off [33,34]. The results indicated that both Ni and Mo play an important role in improving the mechanical properties of the investigated AlCoCrFeNi and CoCrCuFeNi HEAs.

4. Discussion

As mentioned in the previous section, an HEA's mechanical properties are strongly dependent upon their microstructure. Fig. 7 graphically described the relationship between the Ni/Mo content and the volume fractions for FCC/BCC phases, and it further implies a correlation between the VEC and the BCC/FCC phase volume fraction. In other words, "the smaller the VEC, the greater the volume fraction of the BCC phase is." High VEC lead to greater atomic bonding forces, so that atoms tend to arrangement themselves into the FCC structure, which possesses a higher atomic packing density. Meanwhile, low VEC lead to lower atomic bonding forces, where atoms tend to arrangement themselves into the BCC structure denoted by a lower atomic packing density. The valence electron concentration (VEC), which is represented as *e/a*, is also considered

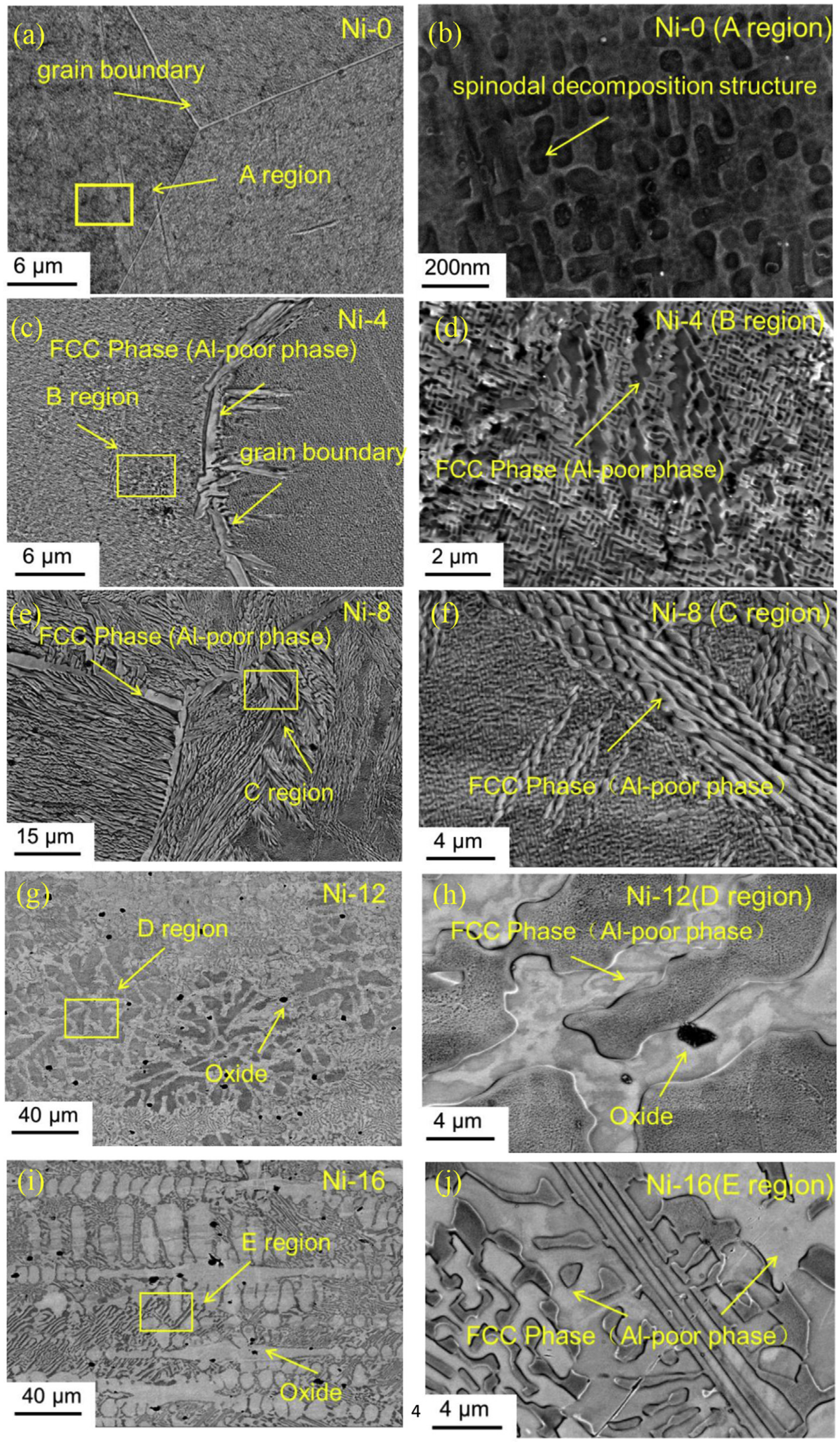

Fig. 4. SEM micrographs of as-cast, $(AlCoCrFeNi)_{100\text{-}x}Ni_x$, where x = 0, 4, 8, 12, and 16. Fig. 4(a, c, e, g, i) are the low magnification images, while Fig. 4(b, d, f, h, j) contain the corresponding high magnification images.

an important parameter for phase structures from an electronic structure perspective. Guo and Liu [10], and Guo et al. [11] proposed that this parameter be used to predict phase formations in HEAs, because it correlates a structure's VEC with its FCC/BCC phase formation. This is because FCC phases are stable at higher VECs (≥ 8), while BCC phases are stable at lower VECs (<6.87), which may be represented generally as $VEC = \sum_{i=1}^{n} C_i (\mathrm{VEC})_i$, where C_i is the atomic percentage, and $(\mathrm{VEC})_i$ is the VEC for the ith element. A

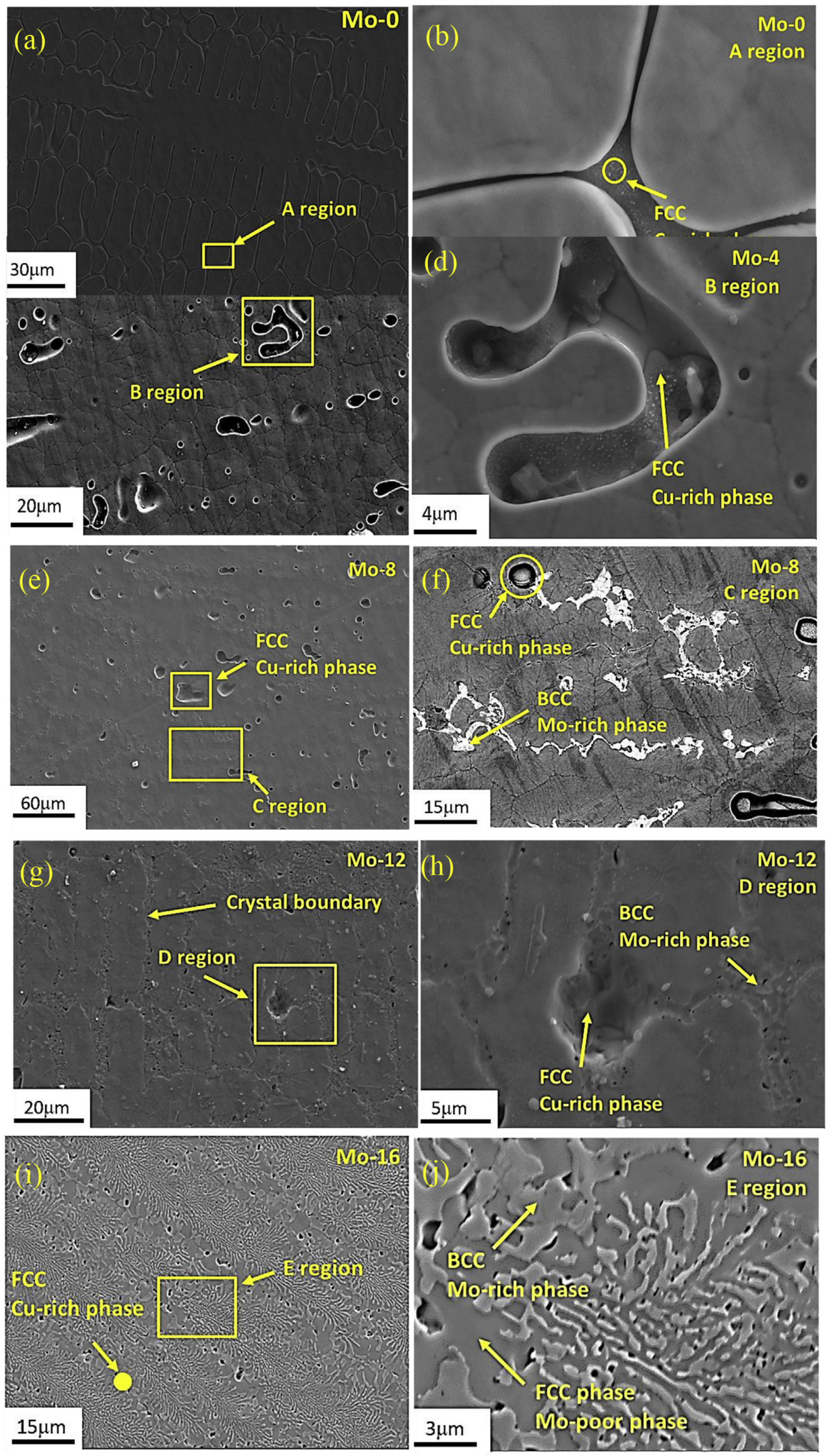

Fig. 5. SEM micrographs of as-cast, $(CoCrCuFeNi)_{100\text{-}x}Mo_x$, where x = 0, 4, 8, 12, and 16. Fig. 5(a, c, e, g, i) are the low magnification images, while Fig. 5(b, d, f, h, j) contain the corresponding high magnification images.

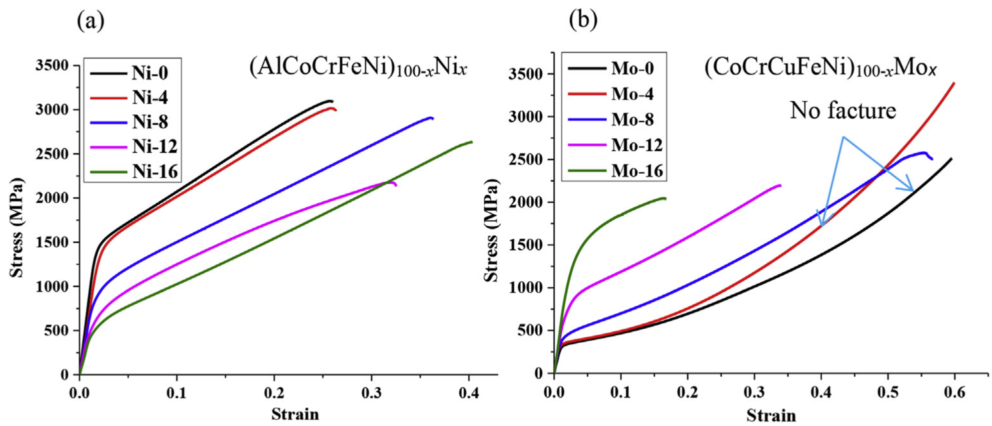

Fig. 6. Compressive mechanical behavior of as-cast, $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ and $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEAs. (a) Engineering stress-strain curve for $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEAs. (b) Engineering stress-strain curve for $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEAs.

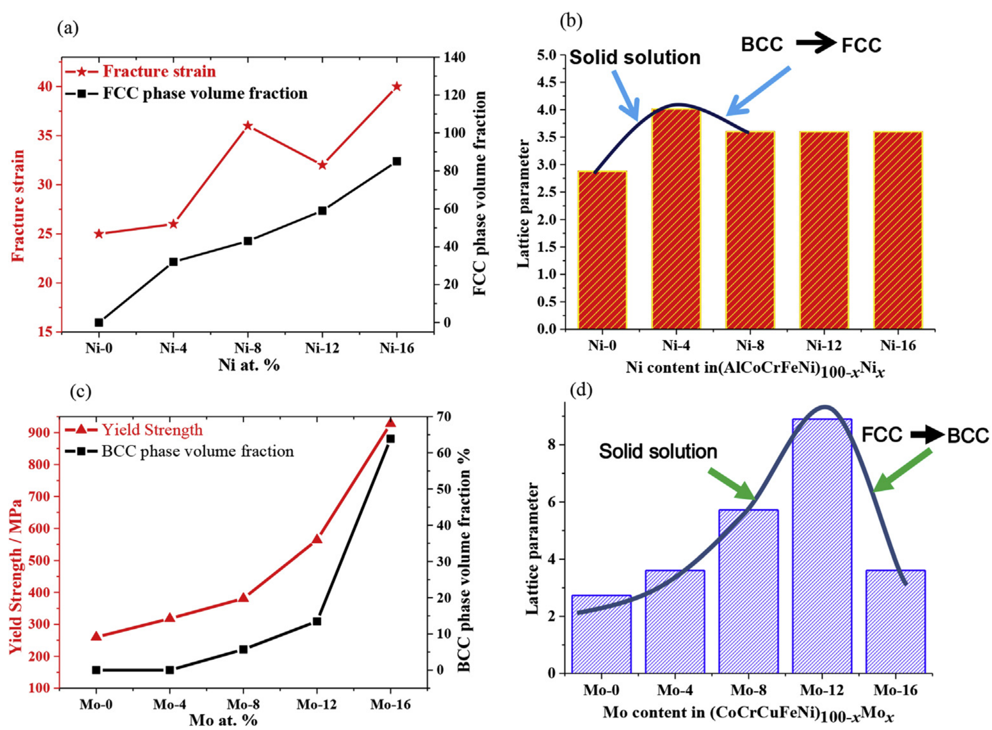

Fig. 7. Mechanical properties for the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ and $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEA systems. (a) Fracture strain and FCC phase fraction with different Ni content in $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEAs. (b) Lattice constants for the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEA system. (c) Yield strength and BCC phase fraction with different Mo content in $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEAs. (d) Lattice constants for the $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEA system.

variety of mechanical properties will be discussed prior to returning to the VEC.

Fig. 7(a) shows the relationship between the fracture strain, the Ni content, and the crystal's volume fractions found in the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEA system. These results, while in an FCC phase, demonstrated that the volume fraction increased with increasing Ni content, which was consistent with the increasing fracture strain as the volume fraction increased. Fig. 7(b) shows the lattice constants for the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEA system. The lattice constants increased when the Ni content increased from 0 to 4 at.%, but they were reduced when the Ni content increased beyond a concentration of 4 at.%. The reason behind the lattice constant reduction was that excessive Ni addition in the alloy yielded a large lattice distortion energy that contributed to the crystal's lattice transformation. When the lattice distortion energy was relaxed, it encouraged phase transformations in the HEA system. This simple point of view works only if no intermetallic compounds are formed [12]. On the contrary, when Mo, acting as a BCC stabilizer, was added to the CoCrCuFeNi HEA system, the volume fraction of the BCC phase increased with increasing Mo content, which was consistent with the yield strength increasing as the volume fraction of the BCC phase increased (refer to Fig. 7(c)). Certainly, the lattice parameter also presented a consistent transformation (refer to Fig. 7(d)).

Fig. 8(a) shows the VEC for the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEA: it shows the BCC region (VEC < 6.87), the FCC region (VEC $\geq$ 8), and the combined BCC + FCC region (6.87 < VEC $\leq$ 8) on the graph. The $(CoCrFeMnNi)_{100\text{-}x}Mo_x$ HEA, when x = 0-16 at.%, is located in the

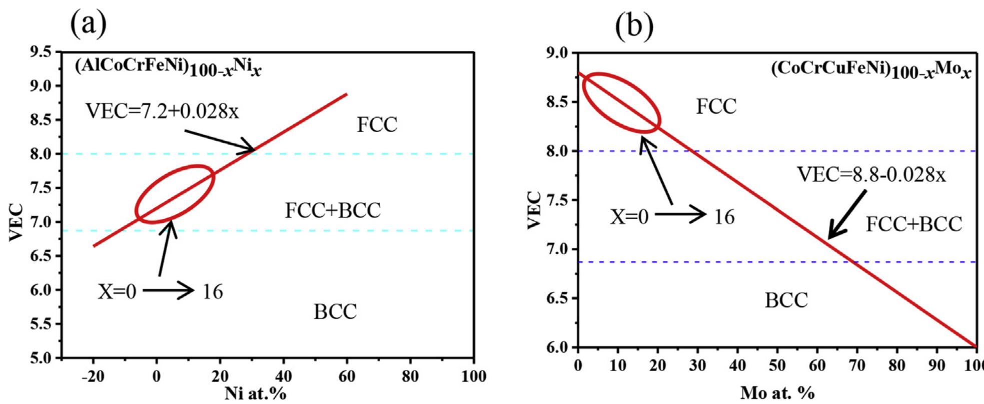

Fig. 8. Valence electron concentration (VEC) for the (a) $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ and the (b) $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEA systems.

BCC + FCC region. The VEC for the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEA system increased, as the Ni content increased, when Ni's VEC ($VEC_{Ni} = 10$) was larger than the average VEC ($VEC_{Ni} = 7.2$) for the AlCoCrFeNi matrix. Conversely, as shown in Fig. 8(b), the VEC for the $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEA, which was imbued with Ni content, increased when Mo's VEC ($VEC_{Ni} = 6$) was lower than the average VEC ($VEC_{Ni} = 8.8$) for the CoCrCuFeNi matrix.

Fig. 9 shows the forming process for the Mo-rich, BCC precipitated phases found in the $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEAs. This process can be divided into three stages. In the first stage, the matrix (CoCrCuFeNi) forms a stable FCC structure as the temperature decreases, this stage is shown in Fig. 9(a). Then the BCC phase stable element (Mo) was incorporated into the FCC matrix when Mo levels are low, this stage is shown in Fig. 9(b-c). When the Mo content exceeds the limits of it can solid solution in the matrix, a portion of the Mo atoms could no longer be incorporated into the matrix, and were then able to form a new phase, the new, Mo-rich, BCC phase continued to increase as the Mo content was increased. At the same time, these atoms (CoCrCuFeNi) integrated in the BCC phase and form BCC Mo-rich phase (see Fig. 9(d-e)). At the end, both of BCC Mo-rich phase and FCC matrix (CoCrCuFeNi) phase contained in the system, this stage is shown in Fig. 9(d-e). This hypothesis was indirectly supported by EDS maps, because of the EDS maps show the BCC phases were Mo-rich (refer to Fig. 3(b)). In a similar way, FCC precipitated phases are also formed in the $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEAs. FCC phase formed in BCC HEA system, has a positive effect in improving the ductility of BCC HEAs. Similarly, BCC phase formed in FCC HEA system, is also beneficial in improving the strength of FCC HEAs.

To further expound upon the mechanism behind the improved HEA's plasticity, a model was proposed in Fig. 10. HEAs with ductile, FCC solid solutions have improved plasticity and lower strength, because they possess more slip directions in their crystal structures [23], while HEAs with hard, BCC/HCP solid solutions demonstrate greater strength and limited plasticity, because this crystal structure contains fewer slip directions [24]. The combined BCC and FCC phases achieved a balance between strength and plasticity. Fig. 10(a) shows the brittle-fracture model for a single phase, BCC HEA, and Fig. 10(b) shows the fracture model for composite, BCC/FCC phases. The latter indicates that the HEA's plasticity can be improved by generating an FCC phase in $(AlCoCrFeNi)_{100\text{-}x}Ni_x$ HEAs. Similarly, the strength in an FCC HEA could be improved by generating BCC phases within the FCC matrix.

5. Conclusions

We introduced a method for designing HEAs based upon the relationship between the VEC and the phase structure. To verify the

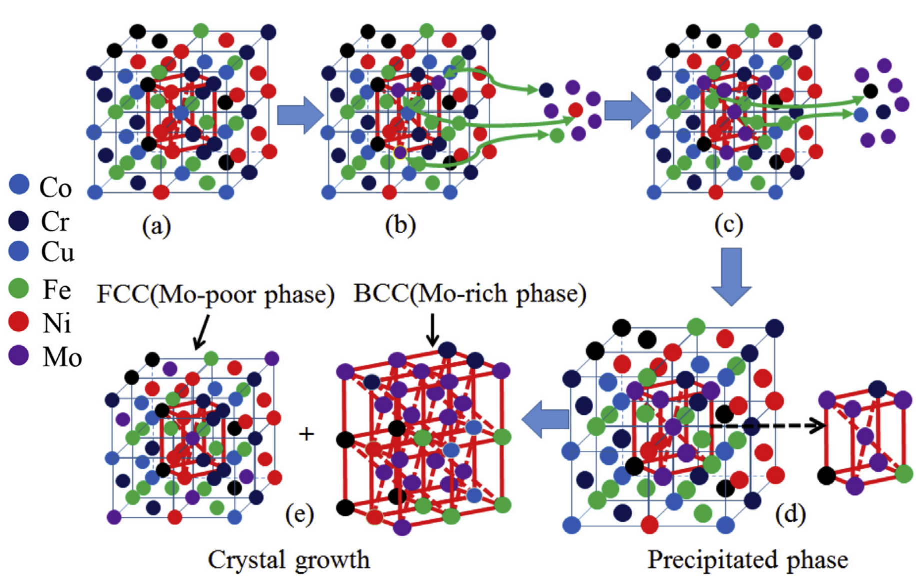

Fig. 9. Forming process for the Mo-rich, BCC precipitated phase in the $(CoCrCuFeNi)_{100\text{-}x}Mo_x$ HEA system.

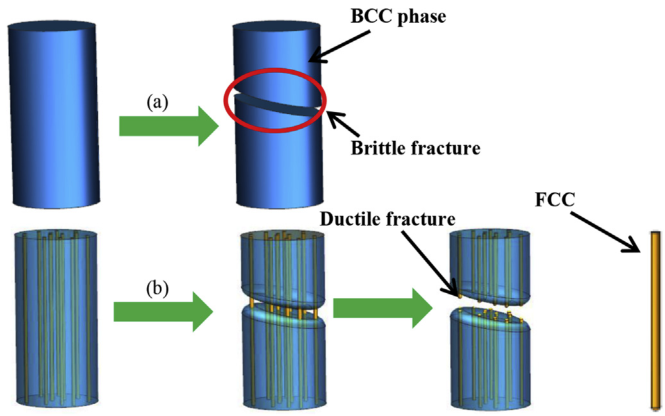

Fig. 10. Fracture model for $(AlCoCrFeNi)_{100-x}Ni_x$ HEAs. (a) Brittle fracture model for a BCC phase matrix. (b) Ductile fracture model for a combined FCC and BCC phase matrix.

validity of this alloy design method, a systematic study on the alloying effects of Mo and Ni were conducted for two common HEAs – $(CoCrCuFeNi)_{100-x}Mo_x$ and $(AlCoCrFeNi)_{100-x}Ni_x$. We emphasize that L-element with low VEC not only promote BCC structure formation, but also promote other structure with low atomic density formation, such as Laves phase, σ phase and so on. According to the experiment and subsequent analysis, we can draw the following conclusions:

1. The phase structure for the $(AlCoCrFeNi)_{100-x}Ni_x$ transformation went from BCC to FCC as the Ni content increased from 0 to 16 at.%, while the FCC volume fraction increased from 0% to 85%, and the VEC increased from 7.2 to 7.6.
2. As the Ni content increased from 0 at.% to 16 at.%, the compressive fracture strain of $(AlCoCrFeNi)_{100-x}Ni_x$ increased from 25% to 40%.
3. The phase structure for the $(CoCrCuFeNi)_{100-x}Mo_x$ matrix transformed from FCC to BCC as the Mo content increased from 0 at.% to 16 at.%, while the BCC volume fraction increased from 0% to 65%, and the VEC decreased from 8.8 to 8.3.
4. As the Mo content increased from 0 at.% to 16 at.%, the compressive yield strength of the $(CoCrCuFeNi)_{100-x}Mo_x$ matrix increased from 260 MPa to 928 MPa.
5. A high VEC is conducive to forming FCC phases that provide improved ductility, while a low VEC is beneficial in forming BCC phases with enhanced strength. Selecting ideal elements, based upon the VEC, is a practical method for designing HEA compositions that balance strength and ductility.

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